TRACE

1 Candidate construction

j: source; b: crop; c: query

$R = (b_R, \Omega_R, c_R)$

4-connected query regions

Top-K by mean confidence

Native 16 x 16 token masks

2 Context probe

Candidate input pixels fixed

Outside: .35 contrast + (g,b,r)

One view per candidate

Original query and tokens

Frozen initial readout

3 Per-image entropy

Prompt/token mean entropy

$\overline{H}_R(\theta)$

Detach; normalize per image

Mean over active images

Visual LayerNorm affine

$$\rho_R = \frac{1}{|\Omega_R|} \sum_{i \in \Omega_R} \pi^0_{b_R, i}(c_R \mid x_R^{\mathrm{ctx}})$$

$$\alpha_{R \mid j} = \mathrm{sg}(\rho_R) \Big/ \sum_{Q \in \mathcal{R}_j^+} \mathrm{sg}(\rho_Q)$$

$$\mathcal{L}_{\mathrm{TRACE}}(\theta) = |\mathcal{I}_+|^{-1} \sum_{j \in \mathcal{I}_+} \sum_{R \in \mathcal{R}_j^+} \alpha_{R \mid j} \overline{H}_R(\theta)$$